Vanessa Oguamanam

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Education

Georgia Institute of Technology - Ph.D. in Computer Science **2017- Present**

- Advised by Dr. Andrea Grimes Parker
- Thesis: *Exploring the Design and Evaluation of AI-Powered Stress-Detection Tools to Combat Inequities in Mental Health Support for Perinatal Black Women*
- *(Anticipated Grad Date)- May 2026*

University of Maryland College Park - M.S. in Human-Computer Interaction **2015-2017**

- Advised by Dr. Tamara Clegg and Dr. Jon Froehlich
- Thesis: HerVillage: Engaging Nigerian Girls in Co-designing an Application to overcome barriers to women's health education

University of Maryland College Park - B.S. in Computer Science **2012-2015**

Publications

(**Authors Contributed Equally to Paper)

JOURNAL PAPERS (UNDER REVIEW)

2026 Sackitey, D.**, **Oguamanam, V.****, Tafete, R., Farinu, O., Carter, S., Idris, M., Hernandez, N., Chandler, R., Parker, A.G. (2026) Reframing Stress Detection Proxies in Social Media Mental Health Prediction for Perinatal Black Women. Nature's **npj Women's Health** *[Submission In-Review]*

2026 **Oguamanam, V.**, Moncur, B., Brown, J., Parker, A.G (2026). Socio-cultural Perspectives on Intent to Use Mobile Apps for Perinatal Wellbeing: A Hybrid NLP Approach. ACM Transactions on Computer-Human Interaction, **ToCHI 2026** *[Resubmission in Progress]*

PEER-REVIEWED CONFERENCE PAPERS

2023 **Oguamanam, V.**, Hernandez, N., Chandler, R., Guillaume, D., McKeever, K., Allen, M., Mohammed, S., & Parker, A. G. (2023, April). An Intersectional Look at Use of and Satisfaction with Digital Mental Health Platforms: A Survey of Perinatal Black Women. In *Proceedings of the 2023 CHI conference on human factors in computing systems*, **CHI 2023**. [Acceptance Rate: 25%]

2020 **Oguamanam, V.**, Cochran, Z., Lee, T., McKlin, T., Abowd, G., & DiSalvo, B. Cultural Clash: Exploring How Studio-Based Pedagogy Impacts Learning for Students in HCI Classrooms. ACM Conference on Designing Interactive Systems, **DIS 2020.** [Acceptance Rate: 25%]

2019 Solomon, A., **Oguamanam, V.**, Guzdial, M., & DiSalvo, B. (2019, July). Making CS Learning Visible: Case Studies on How Visibility of Student Work Supports a Community of Learners in CS Classrooms. In *Proceedings of the 2019 ACM Conference on Innovation and Technology in Computer Science Education*, ***ITiCSE* 2019** (pp. 161-167). [Acceptance Rate: 25%]

2017 Clegg, T., Norooz, L., Kang, S., Byrne, V., Katzen, M., Velez, R., Plane, A.**, **Oguamanam, V.**,

Outing, T., Yip, J., & Bonsignore, E. (2017, May). Live physiological sensing and visualization ecosystems: an activity theory analysis. In *Proceedings of the 2017 Conference on Human Factors in Computing Systems*, ***CHI 2017*** (pp. 2029-2041). [Acceptance Rate: 25%]

2016 Kang, S., Norooz, L., **Oguamanam, V.****, Plane, A.**, Clegg, T. L., & Froehlich, J. E. (2016, June). SharedPhys: Live physiological sensing, whole-body interaction, and large-screen visualizations to support shared inquiry experiences. In *Proceedings of the The 15th International Conference on Interaction Design and Children*, ***IDC 2016*** (pp. 275-287). [Acceptance Rate: 47%]

2016 Norooz, L., Clegg, T. L., Kang, S., Plane, A. C.**, **Oguamanam, V.****, & Froehlich, J. E. (2016). "That's your heart!": Live Physiological Sensing & Visualization Tools for Life-Relevant & Collaborative STEM Learning. Singapore: International Conference of the Learning Sciences, **ICLS 2016**. [Acceptance Rate: 37%]

WORKSHOP PAPERS

2020 **Oguamanam, V.**, & Parker, A.G. (2020, April). Does Race and Culture Matter in the Design of Wearable Health-Tracking Devices. *Position paper that will be presented at the Race in HCI Workshop*, ***CHI 2020.***

2016 **Oguamanam, V.****, & Gansallo, M**. (2016, November). LEDz and Cocktailz. In *Proceedings of the First African Conference on Human Computer Interaction*, **AfriCHI 2016** (pp. 260-262).

Poster Sessions & Symposiums

2026 AIxHealth Research to Impact Collaborative Poster Session (Accepted for April 2026)
2025 GT's Foley Scholar Award Winners' Research Presentation
2024 GT's Doctoral Consortium on Responsible Computing, AI, and Society Inaugural Summit
2024 GT's CHIH Research Award Winners' Recipient Luncheon
2023 Research, and Innovation Development Conference Poster Competition Session
2023 VentureLab Career, Research, and Innovation Development Conference Poster Session
2023 CRA GradCohort Workshop for Underrepresented Minorities Lightning Talk Presentation
2019 Facebook/Meta's Women in Research Symposium Presentation
2017 UMD's Information Studies' Symposium Presentation

Work Experience

USER EXPERIENCE - RESEARCH, DESIGN, DEVELOPMENT

Wellness Technology Lab @ Georgia Tech, Maternal Ease Project **2020 - Present**
Graduate Research Assistantship

- Leading end-to-end user research and evaluation of an LLM-powered stress management system for new moms, integrating qualitative methods (design workshops and surveys), computational analysis, and expert human evaluation to improve model robustness, user trust, and readiness for deployment in a user-facing MVP [In-progress].

- Leading the commercialization of dissertation research insights through participation in pre-seed start-up challenge program with an opportunity to compete in pitch competition for $150K in investment to validate product market fit [In-progress]

- Identified new behavioral and linguistic proxies for mental health prediction contexts in social media through statistical analysis and fine-tuning a domain-adapted RoBERTa model capable of detecting coping-behavior signals beyond emotional-language cues.

- Pioneered mixed-methods survey design study (n = 101) to explore user satisfaction and engagement with consumer mental health apps for new mothers. Leveraged quantitative analysis (descriptive, ANOVA) to validate behavioral and attitudinal trend

Microsoft Research, RiSE Group **Summer 2021**
User Research Intern

- Led a mixed-methods study combining online focus groups (n=41) with hybrid NLP analysis of 5,000+ app reviews to identify alignment and gaps between market-level insights and target users' preferences and contextual factors influencing app selection.

Google, Android/Wear OS Team **Summer 2019**
User Research Intern

- Uncovered critical customer pain points across leading smartwatches through three mixed-method competitive benchmark studies utilizing moderated interviews, surveys, diary studies, and behavioral log analysis to assess product strengths, weaknesses, and support for health-related behaviors.

- Influenced product direction for future iterations of Fitbit and Pixel Watch by synthesizing insights and delivering four recommendations to 150+ cross-functional stakeholders to improve customer adoption and overall health journey support.

Culture and Technology Lab @ Georgia Tech, MoLE Project **Fall 2018-Spr2019**
Graduate Research Assistantship

- Maker Oriented Learning (MoLE) in CS Classrooms.
- Faculty: Dr. Betsy DiSalvo
- Methods: Grounded Theory Inspired Methods (Open Coding, Axial Coding), Thematic Analysis

Technology and Design For Empowerment Lab @ Georgia Tech **Spring 2018**
Independent Study - Mini Research Project

- Near Peer Mentoring in the Global and Local South Project
- Sister Rise Up Project - Georgia Tech High School Peer Mentoring Program
    - Faculty: Dr. Neha Kumar and Dr. Ellen Zegura, and Dr. Barbara Ericson
    - Methods: Semi-Structured Interviews, Open Coding, Thematic Analysis, Focus Groups, Semi-structured Interviews, Thematic Analysis

Georgia Tech - Human-Centered Computing Intro Course **Fall 2017**
Independent Study - Mini Research Project

- Mini Project 1: MyPath Application

    - Advisor: Dr. Beth Mynatt, Mentor: Dr. Maia Jacobs
    - Methods: User Log Data Analysis

Human-Computer Interaction Lab @ University of Maryland College Park **2015-2017**
Graduate Research Assistant

- Advisors: Dr. Jon Froehlich, Dr. Tamara Clegg
- BodyVIS Project
    - Methods: Focus Groups, Co-design, User studies, thematic analysis, prototyping
    - Collaborated on Bodyvis research project, focused on designing and implementing wearable e-textile prototypes with real-time physiological sensing and visualization used to engage and motivate children in learning about the human body
    - Assisted in running user studies and focus groups with formal and informal educators and children to gather data on how Bodyvis can be designed to support STEM learning
    - Coded, transcribed, and synthesized data collected from user sessions to aid in writing and submitting papers to top-tier HCI and Learning Sciences conferences

- SharedPhys Project
    - Method: Field Studies, Mixed-Methods, Pre and Post Questionnaires, Thematic Analysis
    - Co-facilitated 3 back to back field studies
    - Effectively took notes during pilot studies and provided feedback on the visual designs learning activities, and study scripts for upcoming field studies
    - Collected, coded, and analyzed 160 pre and post questionnaires based on a code scheme and transcribed interviews that were used to report on findings for the study

Google, Google Cloud Platform/Cloud Launcher Team **Summer 2016**
User Research Intern

- Resolved 3 critical usability issues and shaped product direction for core Google Cloud Platform workflows by synthesizing insights from design workshops and user studies with B2B stakeholders and internal Googlers.

Human-Computer Interaction Lab @ University of Maryland College Park **Fall 2015**
Undergraduate Research Assistant**,**

- Faculty Lead: Dr. Jennifer Golbeck
- Trolling Comments Detection Project
    - Methods: Web Scraping, Data Analytics, Thematic Coding
    - Explored the best ways possible to detect trolling comments in the comments section online. Web scraped comments from YouTube and Facebook and pulled comments using Twitter API and Perl Module.
    - Collected 3000 comments using one of the methods mentioned above and coded them as being “trolling” or “non-trolling”.
    - Effectively communicated and reported on findings and synthesized them into themes

Inclusive Technology Lab @ Northwestern University **Summer 2014**
Undergraduate Research Intern

- Assigned Advisors: Dr. Anne Marie Piper

- PhD Lead: Dr. Robin Brewer
    - Methods: Interviews, Transcribing, Open Coding, Field Deployment, Prototyping
    - Performed qualitative data collection and analysis to understand older adult social communication needs and practices
    - Developed an ambient art display for late life social communication using Arduino LilyPad, sensors, Arduino Yun, and XBee Module

SOFTWARE DEVELOPMENT EXPERIENCE

The Boeing Company, Commercial Aviation Services, Boeing Online Documents **Summer 2013**
Software Engineer Intern

- Developed new interfaces for our clients' digital operations, repair, & parts manual using C/C++

The Boeing Company, Defense, Space, & Security/Iridium **Summer 2012**
Software Engineer Intern

- Redesigned the system's interface as well as added new functionality to the team's internal system using Java's GUI and libraries

Teaching Experience

TEACHING ASSISTANT

- Teaching Assistant -- Georgia Institute of Technology **Spring 2026**
    - CS6435 Graduate Course, Digital Health Equity, 150 students
    - Instructor: Dr. Andrea Parker

- Head Teaching Assistant -- Georgia Institute of Technology **Summer 2025**
    - CS6435 Graduate Course, Digital Health Equity, 96 students
    - Instructor: Dr. Andrea Parker

- Head Teaching Assistant -- Georgia Institute of Technology **Spring 2025**
    - CS6435 Graduate Course, Digital Health Equity, 92 students
    - Instructor: Dr. Andrea Parker

- Co-Lead Teaching Assistant -- Georgia Institute of Technology **Summer 2023**
    - CS6435 Graduate Course, Digital Health Equity, 18 students
    - Instructor: Dr. Andrea Parker

- Co-Lead Teaching Assistant -- Georgia Institute of Technology **Fall 2022**
    - CS7001 Graduate Course, Grad Studies Computing, 59 students
    - Instructor: Dr. Andrea Parker and Dr. Milos Prvulovic

- Lead Teaching Assistant -- Georgia Institute of Technology **Summer 2022**
    - Graduate Course, Introduction to Cognitive Science, 30 students
    - Instructor: Dr. Michael Helms

- Lead Teaching Assistant -- Georgia Institute of Technology **Spring 2019**
    - CS4803 Undergraduate Course, Big Data and Society, 48 students
    - Instructor: Dr. Ayanna Howard

- Lead Teaching Assistant -- Georgia Institute of Technology **Fall 2018**
    - CS6455 Undergrad/Grad Course, Qualitative Research Methods, 40 students
    - Instructor: Dr. Betsy DiSalvo

- Co-Lead Teaching Assistant -- Georgia Institute of Technology **Spring 2018**
    - CS3790 Undergraduate Course, Introduction to Cognitive Science, 125 students
    - Instructor: Dr. Rosa Arriaga

- Co-Lead Teaching Assistant -- Georgia Institute of Technology **Fall 2017**
    - CS6753 HCI Master's Course, Professional Preparation and Practice -- 100+ students
    - Instructor: Dr. Richard Henneman

INSTRUCTOR

- STEM GEMS -- Georgia Institute of Technology **Summer 2024**
    - Led 50 9th and 10th grade girls together for hands-on activities, career exploration, mindful conversations in Science, Technology, Engineering, and Mathematics

- Georgia State University: Digital Leaders to Learners **Summer 2018**
    - Led a class of 10 middle school and high school students in fundamental concepts in computer science education

- Girls Who Code, Inc. -- Washington, D.C. **Summer 2017**
    - Trained 22 high school girls in Scratch, Python, Arduino C, Django, and web programming and empowered them to develop personally relevant applications that addressed issues in their community

Honors & Awards

- **2026** Institute-wide Outstanding Online Head TA Award
- **2026** Departmental Outstanding Online Head TA Award
- **2024** Foley Scholar PhD Finalist and Winner **(Highest Departmental Student Award)**
- **2023** CHIH Graduate Student Research Publication Award
- **2023** CRA GradCohort Workshop for Underrepresented Minorities & ppw Disabilities
- **2023** College of Engineering- CEED- GradRise Scholarship
- **2022** Jessie & Ralph Pries Fellow Scholarship
- **2022** P.E.O. International Scholarship Award
- **2019** Facebook Emerging Scholars Fellowship
- **2019** CRA GradCohort Workshop for Underrepresented Minorities & ppw Disabilities
- **2018** CRA GradCohort Workshop for Underrepresented Minorities & ppw Disabilities
- **2018** ACM Grant for CHIMe '18 Symposium at CHI
- **2018** Georgia Tech's Sponsorship for Tapia Conference
- **2017** ACM SIGCHI EC Development Fund Grant for Summer/Winter Schools

- **2017** Georgia Tech's Sponsorship for Tapia Conference
- **2017** President's Fellow, Georgia Institute of Technology
- **2017** National Science Foundation Graduate Research Fellowship Honorable Mention
- **2015** National Science Foundation Bridge to the Doctorate Fellowship
- **2015** Maryland Delegate Scholarship
- **2015** The National GEM Consortium Associate Fellow
- **2015** Association of Women in Computing Sponsorship for Grace Hopper Conference
- **2015** Google Sponsorship for Grace Hopper Conference Scholarship
- **2014** Association of Women in Computing Sponsorship for Grace Hopper Conference
- **2014** Team awarded 1st place in the Google API Category at NSBE Hackathon
- **2014** Clifford & Camille Kendall CMNS Scholarship Award
- **2012** National Society of Black Engineers Academic Achievement Award

Mentorship

2025-Present Roshini Deva - Ph.D. Student in Human-Computer Interaction, Georgia Tech
2025-Present Olufunmilola Obielodan - M.S. Human-Computer Interaction, Georgia Tech
2025-Present Monneh Sherman - Undergraduate Student in Neuroscience, Georgia Tech
2025-Present Griffin Bryant - High School Student, The Forest School
2024-2025 Jordan Brown - M.S. Human-Computer Interaction, Georgia Tech
2024-2025 Binta Moncur - M.S. Human-Computer Interaction, Georgia Tech
2021-2022 Kai McKeever - Undergraduate Student in Computer Science, Georgia Tech
2021-2022 Morgan Allen - Undergraduate Student in Pre-Med, Spelman College
2021-2022 Sabreen Mohammed - M.S. in Public Health, Emory University
2021-2022 Dominique Guillaume - PhD Student in Nursing, John Hopkins University

Service

OUTREACH, ORGANIZATIONS, & MEMBERSHIPS

- **2026** **Participant -** Klaus Start-up Challenge Accelerator Program
- **2024** **Mentor -** Indigo Research
- **2024** **Committee Member –** Georgia Chapter of Postpartum Support International
- **2023** **Member** - VentureLab Female Founders Cohort
- **2023** **Mentor -** Girls Inc. of Greater Atlanta
- **2022** **Research Mentor --** Ceed's Summer Undergraduate Research in Engineering Program
- **2020** **Member -** African Graduate Students Connect
- **2020** **Member -** Black Graduate Students Association at Georgia Tech
- **2019** **Programs Committee --** National Society of Black Engineers (GTSBE Chap)
- **2019** **Communications Chair, WebMaster --** African Graduate Students Connect @ GT
- **2018** **Communications Chair, WebMaster --** African Graduate Students Connect @ GT
- **2018** **Member --** Diversifying Future Leadership in the Professoriate (FLIP) Alliance
- **2017** **Volunteer-**Working to Advance STEM Education for African Women Foundation
- **2017** **Co-Founder/Technical Project Manager** - She Hacks Africa
- **2015** **Volunteer** - Louis Stokes Alliances for Minority Participation STEM Expo @ UMD
- **2014** **Tech Assistant** - Black Girls Code
- **2014** **Session Lead Instructor --** Annual Cyber Security Workshop for Middle School Girls

- **2013 Session Lead Instructor --** Annual Cyber Security Workshop for Middle School Girls
- **2012 Session Lead Instructor --** Annual Cyber Security Workshop for Middle School Girls
- **2012 - Volunteer/Mentor --** Microsoft Digigurlz Camp

ACADEMIC SERVICE

Student Volunteer

2026 National Society of Black Engineers National Convention
2023 Black is Tech Conference
2018 Black Compute HER Conference
2016 AfriCHI First Inaugural Conference

Peer Reviewer - Conference/Journals Papers

2025 JMIR - Full Paper
2025 CHI - Full Paper
2019 FABLearn - Short Paper
2018 ICLS - Paper Poster

Panelist

2021 Microsoft Research Internship Panelist
2018 Georgia Tech's FOCUS Grad Recruitment Program Panelist